

1. IPC using Shared Memory

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>
#include <unistd.h>

int main() {
    int shmid;
    char *str;

    // 1. Set size
    int size = 1024;

    // 2. Allocate shared memory
    shmid = shmget((key_t)1234, size, 0666 | IPC_CREAT);

    if (shmid < 0) {
        printf("Shared memory creation failed\n");
        return 1;
    }

    // 3. Attach
    str = (char*) shmat(shmid, NULL, 0);

    // 4. Write data
    sprintf(str, "Hello from Sender Process!");

    printf("Data written to shared memory: %s\n", str);

    // 5. Delay
    sleep(10);

    // 6. Detach
    shmdt(str);

    return 0;
}
```

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main() {
    int shmid;
    char *str;

    // 1. Set size
    int size = 1024;

    // 2. Access shared memory
    shmid = shmget((key_t)1234, size, 0666);

    if (shmid < 0) {
        printf("Shared memory access failed\n");
        return 1;
    }

    // 3. Attach
    str = (char*) shmat(shmid, NULL, 0);

    // 4. Read data
    printf("Data received from shared memory: %s\n", str);

    // 5. Detach
    shmdt(str);

    return 0;
}
```

Sender Output:

Data written to shared memory: Hello from Sender Process!

Receiver Output:

Data received from shared memory: Hello from Sender Process!